Saiyed Shizain | CU24250233 | Roll: 48
BTech CSE (A) | 3rd Yr - 5th Sem | 2024-2028

Full-Stack Web Development — BTCS303P

LAB SHEET — 1: Core Web Structures Using Semantic HTML

Task 1.1 — Structured HTML resume with semantic layout blocks

Aim. Design a resume page using header, section and footer blocks conforming to modern accessibility parameters (landmarks, ARIA labels, alt/associated text).

Theory. Semantic landmarks let assistive tech navigate; role/aria-labelledby attributes name each region explicitly.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Resume of Saiyed Shizain, BTech CSE student.">
  <title>Resume — Saiyed Shizain</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 0; background: #f4f6f8; color: #222;
}
    header, footer { background: #1f2937; color: #fff; padding: 24px 7%; }
    main { max-width: 800px; margin: auto; padding: 24px 7%; background: #fff; }
    section { margin-bottom: 24px; }
    h1 { margin: 0 0 4px; }
    a { color: #2563eb; }
    ul { line-height: 1.7; }
  </style>
</head>
<body>
  <header role="banner">
    <h1>Saiyed Shizain</h1>
    <p>BTech Computer Science &amp; Engineering (A) &mdash; 3rd Year, 5th Semester</p>
    <p>CU ID: CU24250233 &middot; Roll No: 48 &middot;
      <a href="mailto:saiyed.shizain@example.com" style="color:#93c5fd">saiyed.shizain@e
xample.com</a>
    </p>
  </header>
  <main role="main">
    <section aria-labelledby="edu">
      <h2 id="edu">Education</h2>
      <p><strong>BTech CSE</strong>, 2024&ndash;2028. Relevant coursework:
        Full-Stack Web Development (BTCS303P), Computer Vision and Digital Image Process
ing.</p>
    </section>
    <section aria-labelledby="skills">
      <h2 id="skills">Technical Skills</h2>
      <ul>
        <li>HTML5, CSS3, JavaScript</li>
        <li>Node.js, Express.js, React, MongoDB</li>
        <li>Python, OpenCV, scikit-learn</li>
        <li>Git &amp; GitHub, Docker basics</li>
      </ul>
    </section>
    <section aria-labelledby="projects">
      <h2 id="projects">Projects</h2>
      <ul>
        <li><strong>CampusConnect</strong> &mdash; role-based event portal (MERN + Socke
t.io + Redis).</li>
        <li><strong>CV &amp; DIP lab sheets</strong> &mdash; executable Jupyter notebook
s with print-ready PDFs.</li>
      </ul>
```

```
    </section>
  </main>
  <footer role="contentinfo">
    <p>&copy; 2026 Saiyed Shizain &mdash; Lab Sheet 1, Task 1.1</p>
  </footer>
</body>
</html>
```

```
$ python -m http.server 8811
$ curl -s -o /dev/null -w '%{http_code}' http://127.0.0.1:8811/task-1.1-resume.html
200
[structure check] Structure check: header/section/footer/h1 present; mailto + description meta
present.
```

Conclusion. Page serves correctly and meets the task's structural requirements. — Saiyed Shizain

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Task 1.2 — Nested navigation menu with absolute-path external links

Aim. Build a nav bar from nested list anchors; internal links are relative, external resources use absolute URLs; submenu opens on hover and keyboard focus-within.

Theory. Nested ul/li models hierarchy; :focus-within keeps it keyboard-accessible, unlike hover-only menus.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Nested Navigation Menu — Saiyed Shizain</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 0; background: #f4f6f8; }
    nav { background: #1f2937; padding: 14px 7%; }
    nav ul { list-style: none; margin: 0; padding: 0; display: flex; gap: 4px; flex-wrap
: wrap; }
    nav li { position: relative; }
    nav a { display: block; color: #e5e7eb; text-decoration: none; padding: 10px 14px; b
order-radius: 6px; }
    nav a:hover, nav a:focus { background: #374151; color: #fff; }
    nav ul ul { display: none; position: absolute; top: 100%; left: 0; background: #1118
27;
                min-width: 200px; border-radius: 6px; padding: 6px; flex-direction: colu
mn; }
    nav li:hover > ul, nav li:focus-within > ul { display: flex; }
    main { padding: 24px 7%; }
  </style>
</head>
<body>
  <nav aria-label="Main navigation">
    <ul>
      <li><a href="task-1.1-resume.html">Home</a></li>
      <li><a href="task-1.3-catalog.html">Catalog</a>
        <ul>
          <li><a href="task-1.3-catalog.html#html">HTML Guides</a></li>
          <li><a href="task-1.3-catalog.html#css">CSS Guides</a></li>
          <li><a href="task-1.3-catalog.html#js">JS Guides</a></li>
        </ul>
      </li>
      <li><a href="https://developer.mozilla.org/en-US/docs/Web/HTML">MDN HTML (absolute
)</a></li>
      <li><a href="https://nodejs.org/en/docs">Node.js Docs (absolute)</a></li>
    </ul>
  </nav>
  <main>
    <h1>Nested Navigation Menu</h1>
```

```
    <p>By Saiyed Shizain — Lab Sheet 1, Task 1.2. Hover or keyboard-focus
      <strong>Catalog</strong> to reveal the nested submenu. External links use absolute
 URLs.</p>
  </main>
</body>
</html>
```

```
$ python -m http.server 8811
$ curl -s -o /dev/null -w '%{http_code}' http://127.0.0.1:8811/task-1.2-navmenu.html
200
[structure check] Structure check: nav/ul/li/a present; absolute https links present; submenu
list nested inside Catalog item.
```

Conclusion. Page serves correctly and meets the task's structural requirements. — Saiyed Shizain

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Task 1.3 — Itemized catalog with definition lists and metadata terms

Aim. Present guide metadata with structured dl/dt/dd term-description pairs inside labelled sections.

Theory. dl conveys term-description association to screen readers better than plain headings plus paragraphs.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Web catalog of study guides by Saiyed Shizain.">
  <title>Web Catalog — Saiyed Shizain</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 0; background: #f4f6f8; color: #222;
}
    header { background: #1f2937; color: #fff; padding: 24px 7%; }
    main { max-width: 800px; margin: auto; padding: 24px 7%; background: #fff; }
    dl { border: 1px solid #e5e7eb; border-radius: 8px; padding: 16px 20px; margin-botto
m: 20px; }
    dt { font-weight: bold; color: #1d4ed8; margin-top: 10px; }
    dd { margin: 4px 0 10px 16px; line-height: 1.6; }
    footer { text-align: center; padding: 16px; color: #6b7280; }
  </style>
</head>
<body>
  <header>
    <h1>Study Guides Catalog</h1>
    <p>Curated by Saiyed Shizain — Lab Sheet 1, Task 1.3</p>
  </header>
  <main>
    <section id="html" aria-labelledby="h-html">
      <h2 id="h-html">HTML Guides</h2>
      <dl>
        <dt>Semantic HTML5</dt>
        <dd>Using header, nav, main, section and footer landmarks so assistive
          technologies can navigate the page structure.</dd>
        <dt>Forms &amp;amp; Validation</dt>
        <dd>Native input types, required attributes and accessible labels for
          reliable client-side data capture.</dd>
      </dl>
    </section>
    <section id="css" aria-labelledby="h-css">
      <h2 id="h-css">CSS Guides</h2>
      <dl>
        <dt>Flexbox Layout</dt>
        <dd>One-dimensional fluid layouts that wrap gracefully across mobile viewports.<
/dd>
        <dt>CSS Variables</dt>
        <dd>Theme tokens (colors, spacing) overridable at runtime for dark-mode toggling
```

```
.</dd>
      </dl>
    </section>
    <section id="js" aria-labelledby="h-js">
      <h2 id="h-js">JavaScript Guides</h2>
      <dl>
        <dt>DOM Scripting</dt>
        <dd>Creating, appending and removing nodes at runtime with validation error flag
s.</dd>
        <dt>Fetch &amp;amp; REST</dt>
        <dd>Consuming JSON APIs with async functions and rendering the results.</dd>
      </dl>
    </section>
  </main>
  <footer>
    <p>&amp;copy; 2026 Saiyed Shizain</p>
  </footer>
</body>
</html>
```

```
$ python -m http.server 8811
$ curl -s -o /dev/null -w '%{http_code}' http://127.0.0.1:8811/task-1.3-catalog.html
200
[structure check] Structure check: dl/dt/dd present across three labelled sections with
fragment anchors.
```

Conclusion. Page serves correctly and meets the task's structural requirements. — Saiyed Shizain

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